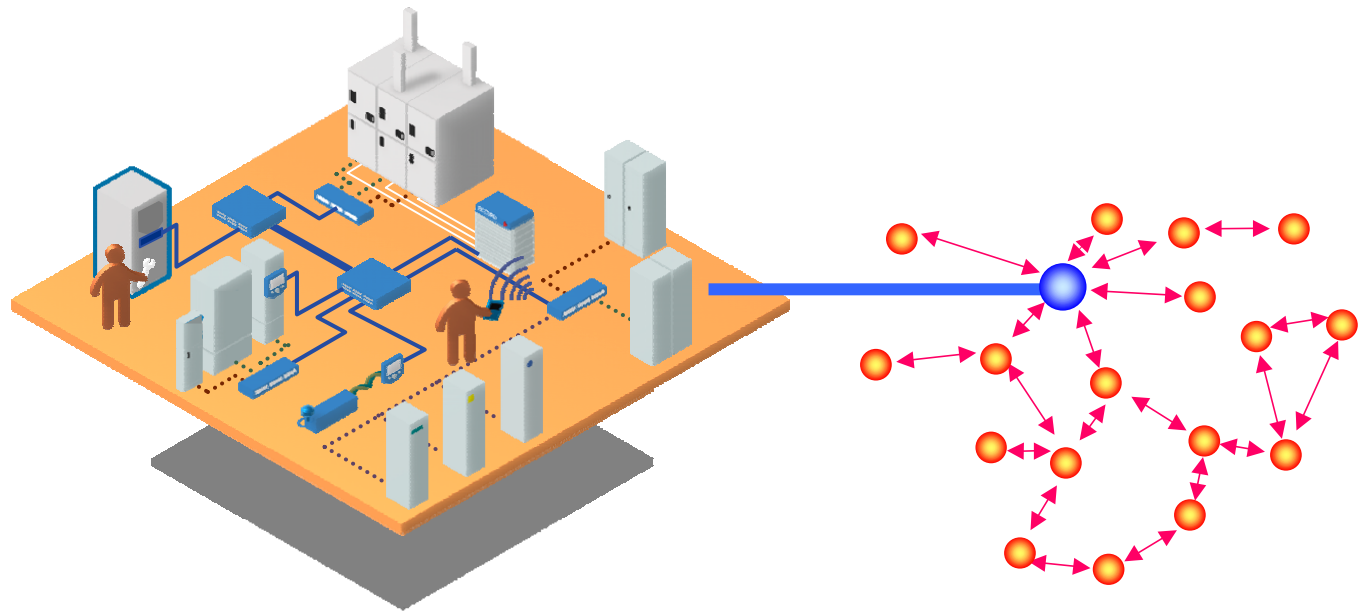




Modeling Wireless Sensor Network Architectures using AADL

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Outline

- Wireless Sensor Networks for Industrial Environments
 - Background & Motivation
 - Architectural analysis needs
- Modeling WSN architectures using AADL
- Proposals WSN specific AADL extensions
- Perspectives

A Different Domain: Energy Management Solutions

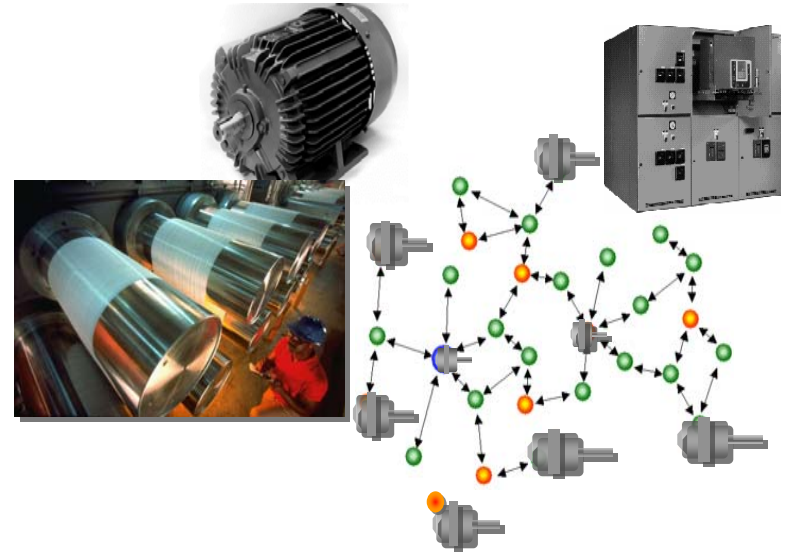
- Electric Motor Driven Process accounts for **23%** of all US electricity sold
- US manufacturing sector: 32,000,000 (**99.6%**) motors rated 500HP or less
- Consumes **71%** of electrical energy used in industrial process plants



Any practical contribution to improve motor efficiency and minimize unscheduled downtime will have a huge impact in energy consumption

Wireless Sensor Networks

- Excellent Proposition for Retrofitting
 - Minimize installation costs
 - Minimum intrusion
 - Self-configuring
- Low Cost
 - Minimizes Wiring costs
 - Is Greener
- Architectural Flexibility
 - Redistribute Sensing, Computation & Communication resources after deployment



What Makes a WSN?

- Small, low-cost multipurpose nodes
 - slow processor speeds (< 10 MHz)
 - low memory (< 10 KB RAM)
 - low bandwidth radio (< 250 kbps)
 - limited battery power (< 4000 mAh)
- Compensated with Redundancy
 - Spatial
 - Time
 - Frequency



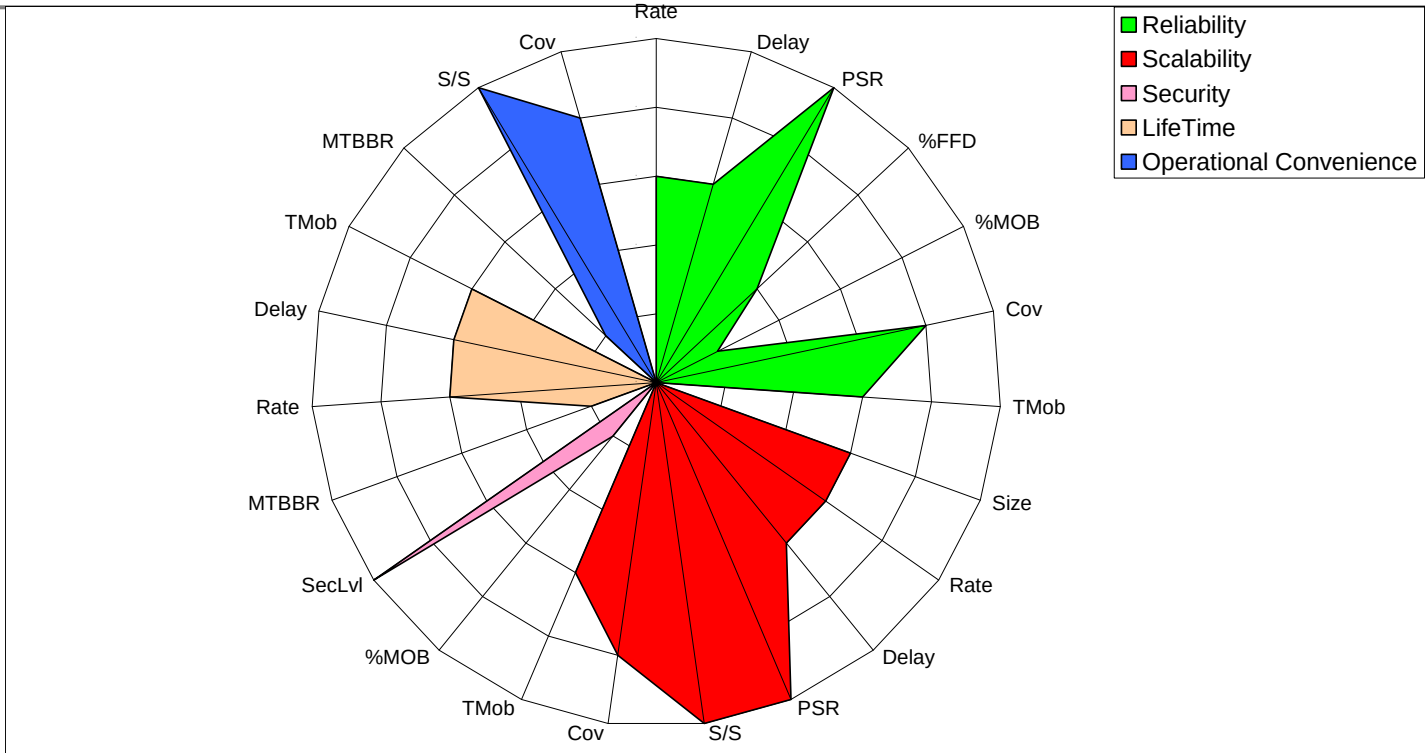
“Face Value” Concerns for WSNs

- Cheap, unreliable hardware
 - failures during assembly, handling and mostly, deployment
- Unreliable system software
 - COTS components: limited or no assumptions about fairness, deadlock freedom, atomicity, etc.
 - Faults because of exceeding node and network capacity
 - slow processor speeds (< 10 MHz)
 - low memory (< 10 KB RAM)
 - low bandwidth radio (< 250 kbps)
 - limited battery power (< 4000 mAh)
- Harsh, changing environments
 - damaged sensors in extreme heat, cold
 - false sensor detections due to wind, rain, etc.

Dependability Concerns for WSNs

- Inadequacy of existing analytical or simulation tools for dependability
 - Current research
 - Process-specific spatial and temporal fault variability
 - e.g. Pump systems vs. load delivery systems
 - Changes reflected in quality of communication channel
 - Triggers real-time topology changes even for non-mobile nodes
 - lots of surprises at scale
- Complexity of system integration
 - array of non-trivial protocols
 - sensing, routing, timesync, localization, etc.
 - hard to reason about correctness – current models not built for this
 - complex dependencies and interactions between protocols
 - hard to guarantee performance
 - co-design and tuning of protocols with resource conflicts
 - COTS components with unknown dependability parameters

Constraints Affecting Dependability



Application Design Space Parameters	Facilities Management	Shorthand	Required Levels			
Network Size	<65K	Size	<3	<256	<65K	<32M
Mean Time Between Battery Replacement/Recharges	<1wk	MTBBR	<1wk	<1mo	<6mo	<1y
Application Data Rate (kbps)	<1	Rate	<0.01	<0.1	<1	<10
End-to-end Delay	<100ms	Delay	<10s	<1s	<100ms	<10ms
Success Rate	>99.9999%	PSR	>90%	>99%	>99.99%	>99.999%
FFD-to-TotalNodes Ratio	<75%	%FFD	=100%	<75%	<50%	<25%
Mobile Nodes-to-TotalNodes Ratio	<1%	%MOB	<1%	<25%	<50%	<75%
Source/Sink Ratio	100	S/S	1/100	1/10	1	10
Coverage (sensors/sq m)	<10	Cov	<1	<2	<5	<10
Mobility (Mean Link Change Time)	1min	TMob	1wk	1h	1min	1s
Security Level	Level 7+	SecLvl	None	Level 0	Level 3	Level 7

Dependability Metrics in WSN

- Ability to deliver a service that can be trusted
- Expressed in traditional metrics
 - MTTF
 - Node failure: MTTF for # of nodes fails
 - Network partition: MTTF for major partition
 - Coverage degradation: MTTF for # of percentage coverage
 - Communication reliability:
 - Latency curves
 - Packet successful rate curve

What if we are successful? -Evolvability

- WSN will Grow in Size and Complexity!
 - Like Internet, users will piggy back on top of the existing WSN for different, unknown apps
 - A safety and security issue sooner or later
 - Incoming stream of research results
 - Efficient routing schemes
 - Energy aware networks
 - Many options to try and test in real settings
- Architecture analysis is critical to assess important quality and dependability attributes.

Why AADL?

- Not only for “Automotive”
 - Set of functionalities specific to model embedded systems
 - If people are planning to use it in planes and cars, why not WSN?
- It works!
 - Extensibility: add your own plug-in
- It is Standard
 - Interoperability
 - Several providers, including Open Source
- Potential Backbone language to exchange domain specific information

Our goals with AADL

(1) InfoBase for Architecture Description

Functional	Informational
Deployment	Operational (*)
Concurrency	

What is my architecture?

Which architecture is better?

Where is the bottleneck?

(2) Architecture comparison

Dependability	Evolution
Development (*)	Cost (*)
Performance	

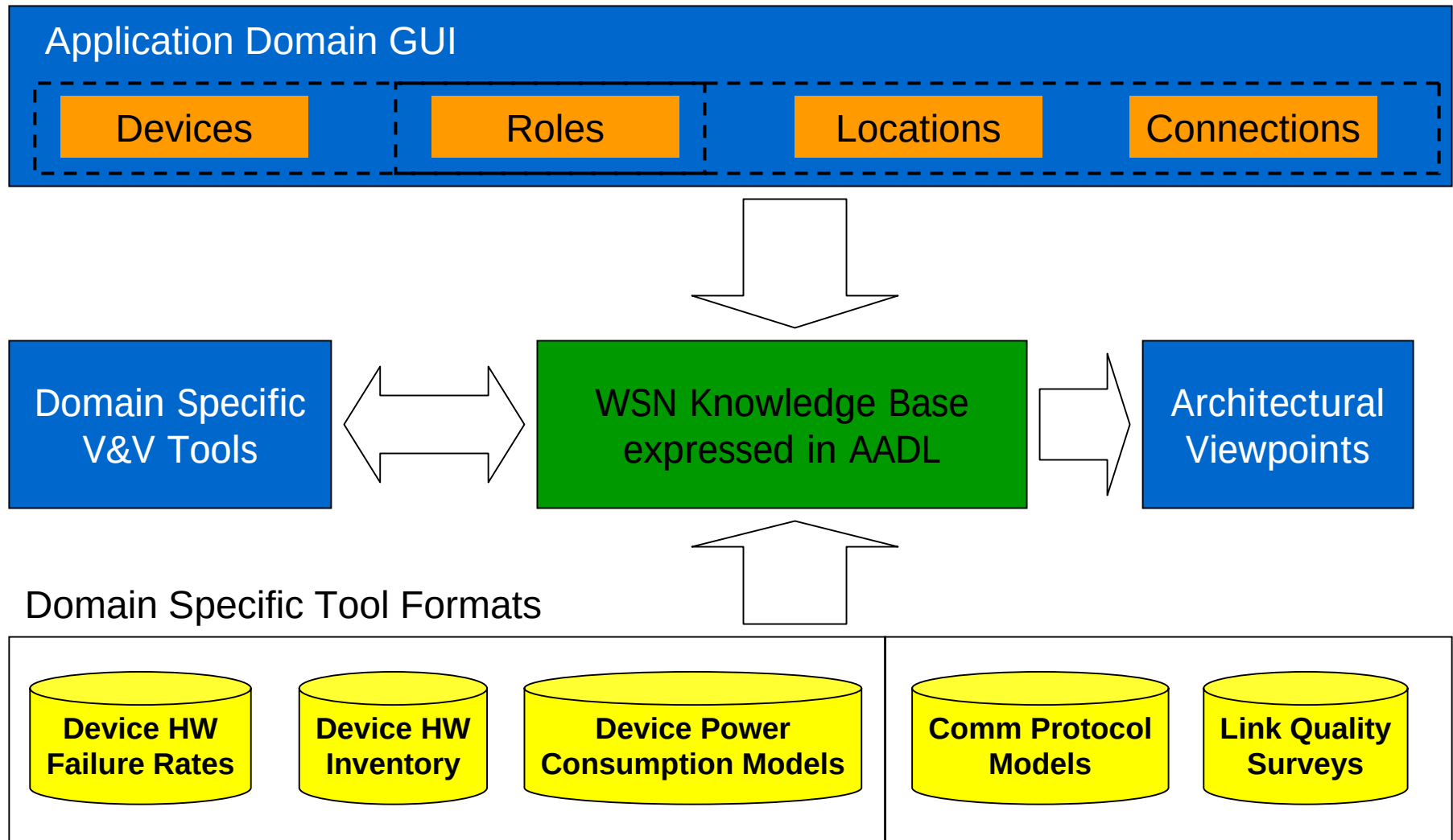
(3) DS Parameter sensibility analysis

End-user tool

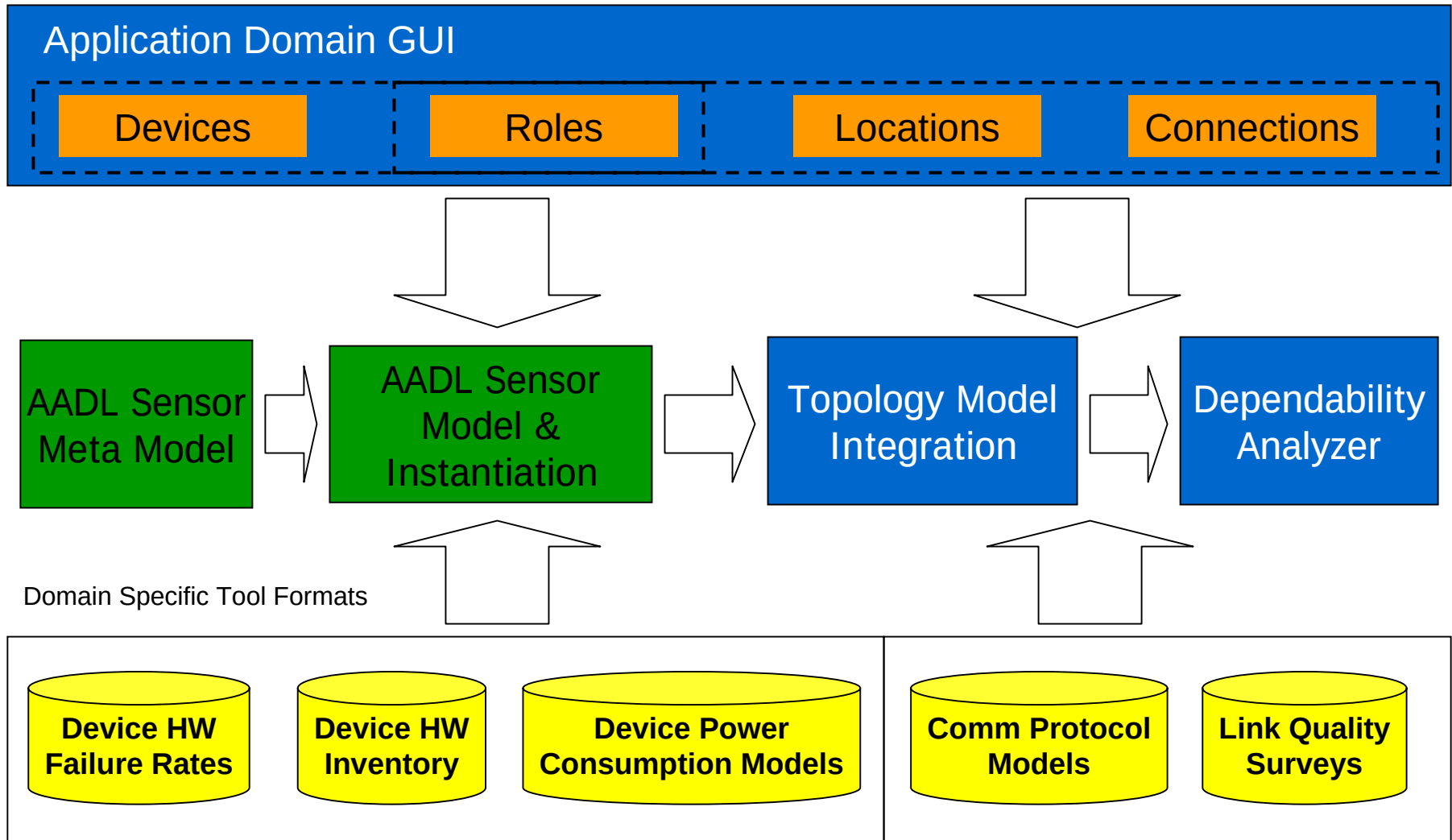
(*) not in AADL's "original" scope



The Vision



Case Study: Where we are now



Case Study: Modeling a Sensor in AADL

AADL Constructs:

- Software category

- Process
- Subprogram
- Data
- Thread
- Thread group



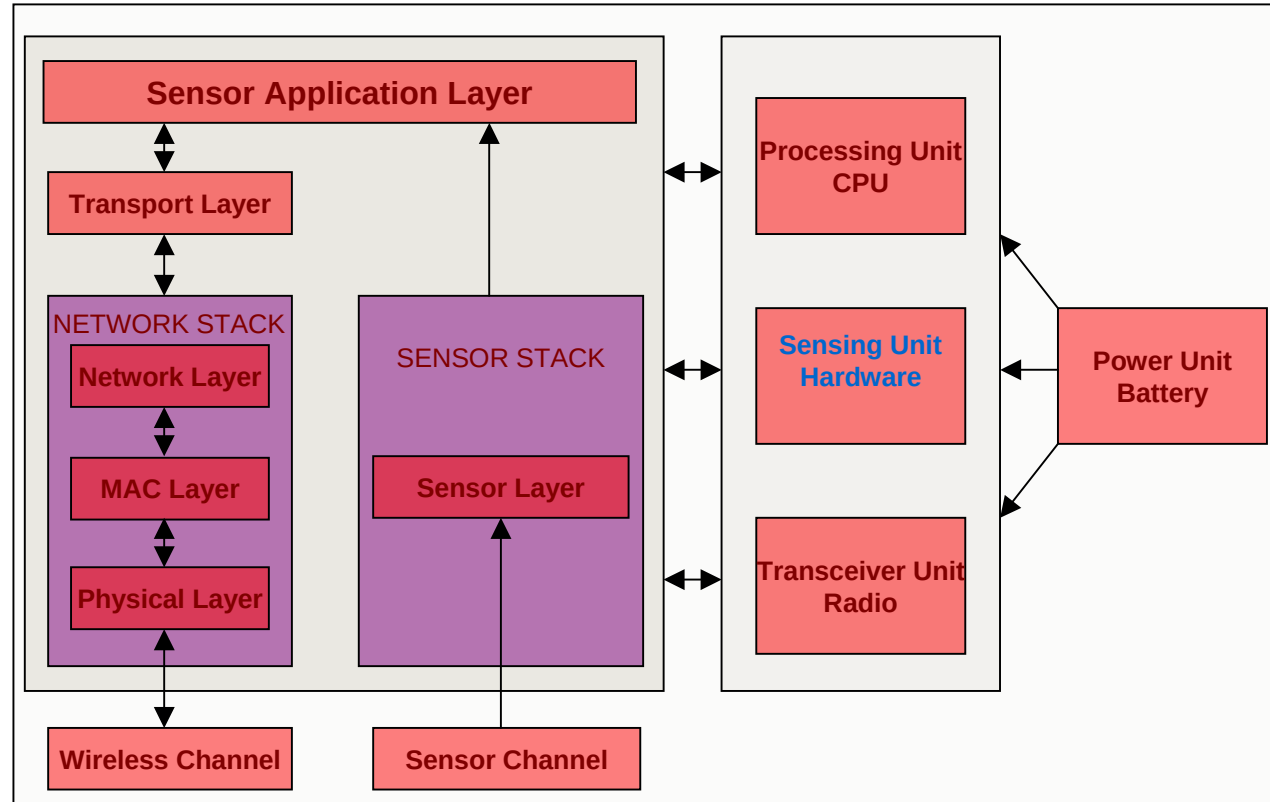
- Platform category

- Processor
- Memory
- Device
- Bus



- Composite

- System



Mapping

- Layers: **Threads**
- Wireless Channel: **Bus**
- Sensor Channel: **Bus**
- Processing Unit: **Processor**
- Sensing Unit: **Device**

- Transceiver Unit: **Device**
- Power Unit: **Device**
- Sensor Measurements: **Data**
- Messages Exchanged: **Data, Event Ports**

AADL Model - Node

- No concept of location, key for WSN analysis
 - Security
 - PHY condition
- Mitigated with AADL property extensions

```
system implementation NODE.NODE0
properties
  --Location Properties
  NodeProp::node_id => 0;
  NodeProp::node_x => 0.0;
  NodeProp::node_y => 0.0;
  --Hardware Components Properties
  NodeProp::node_memory => 1000.0;
  NodeProp::node_cpu_speed => 1000.0;
  --Environments Properties
  NodeProp::op_temperature => 30.0;
  NodeProp::dormrant_temperature => 40.0;
  --Operating Profiles Properties
  NodeProp::humidity => 20.0;
  NodeProp::vibration_level => 0.0;
  NodeProp::case_ID => 0;
  --Battery Properties
  NodeProp::isBattery => 0;
  NodeProp::node_battery_energy => 2100.0;
  --Node state
  NodeProp::node_time => 0.0;
end NODE.NODE0;
```


AADL Model - Network

- No construct to express multiplicity!

system implementation WSN.NLA
subcomponents

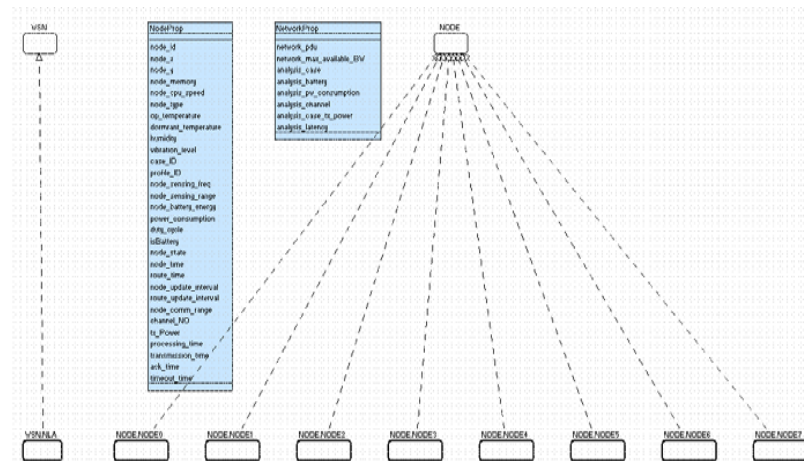
```
NODE0:  system NODE.NODE0;
NODE1:  system NODE.NODE1;
NODE2:  system NODE.NODE2;
NODE3:  system NODE.NODE3;
NODE4:  system NODE.NODE4;
NODE5:  system NODE.NODE5;
NODE6:  system NODE.NODE6;
NODE7:  system NODE.NODE7;
```

properties

```
NetworkProp::network_channel => 13;
```

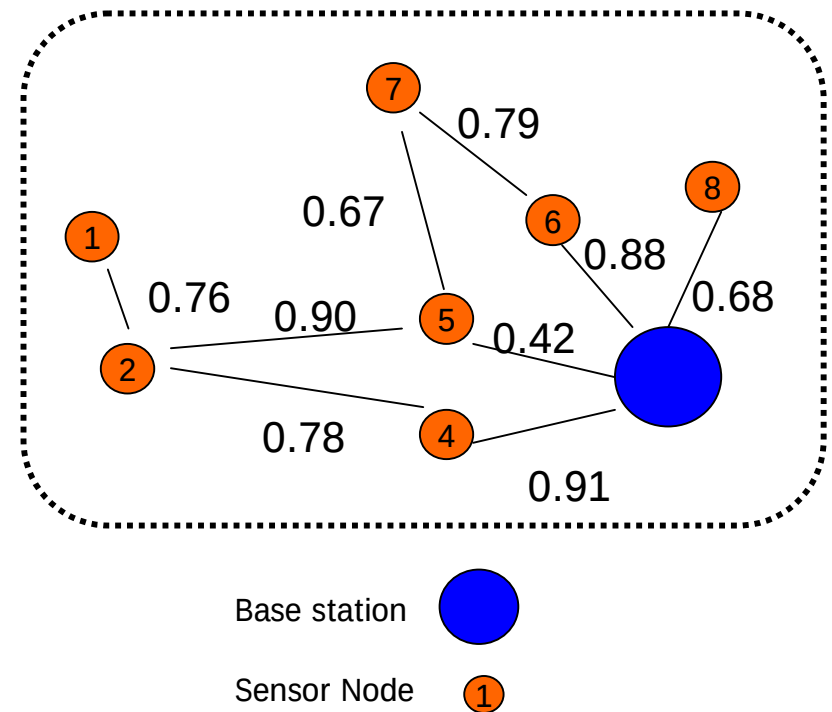
```
NetworkProp::network_GrenadaData=>"\c\dev\psr1.txt"
```

```
end WSN.NLA;
```

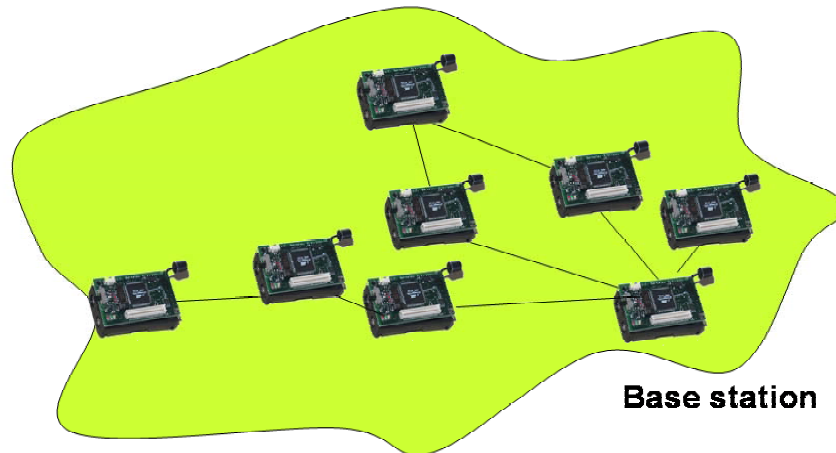


AADL Network - Topology

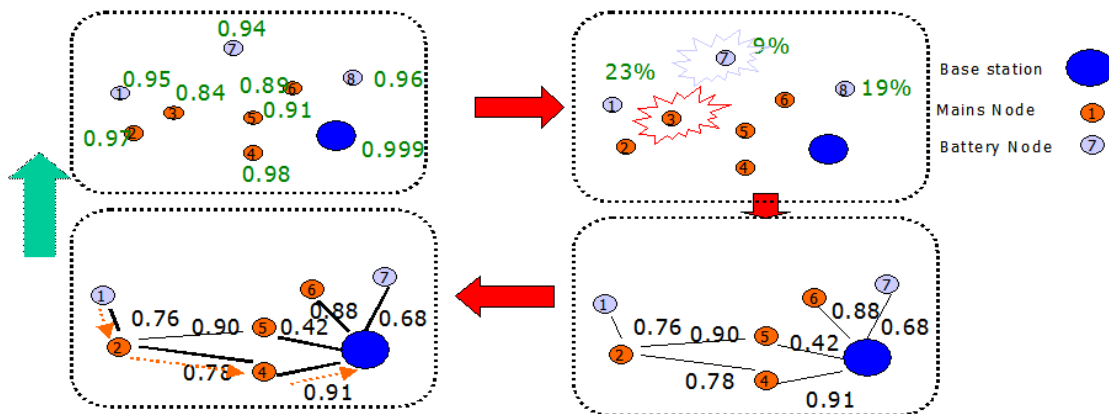
- The bus concept is inadequate
 - Graph concepts hard to do with property extensions
 - Lossy links model
 - Grouping/Multiplicity



Dependability Analyzer Case Study



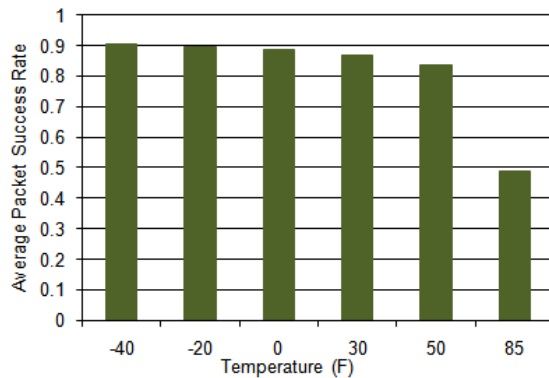
- Network configured in ZigBee “Mesh” mode
- All nodes configured as FFD devices
- AODV routing



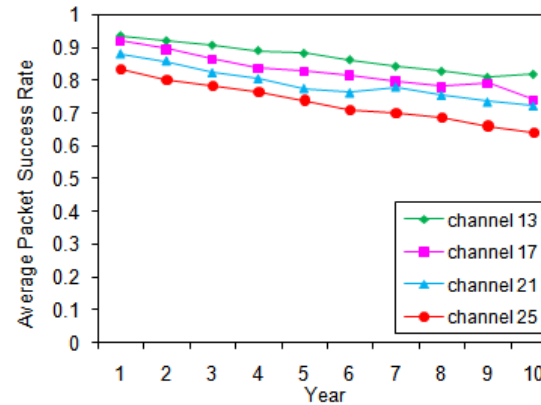
- **Monte Carlo simulation**
 1. Get node aliveness data
 2. Decide whether node is alive or dead by throwing unbiased coin
 3. Query link quality of alive node
 4. Calculate reliability

Analyzer Outputs

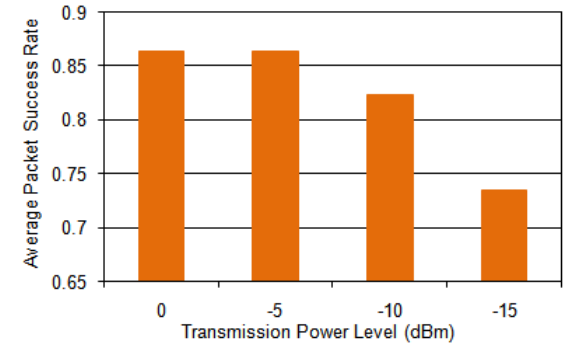
PSR vs. temperature



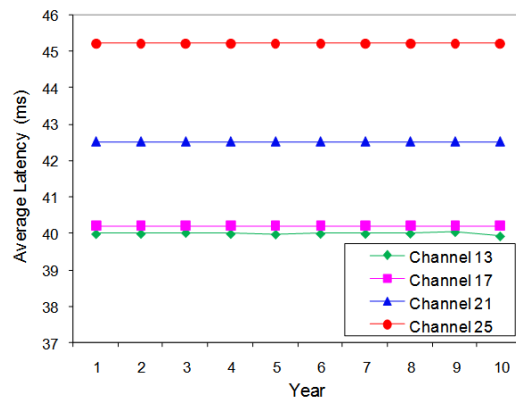
PSR over time vs frequency



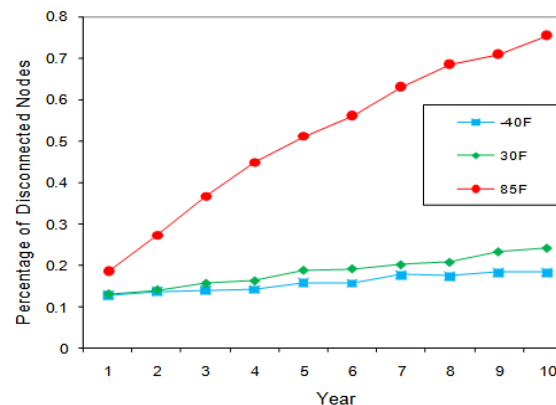
PSR vs. transmission power



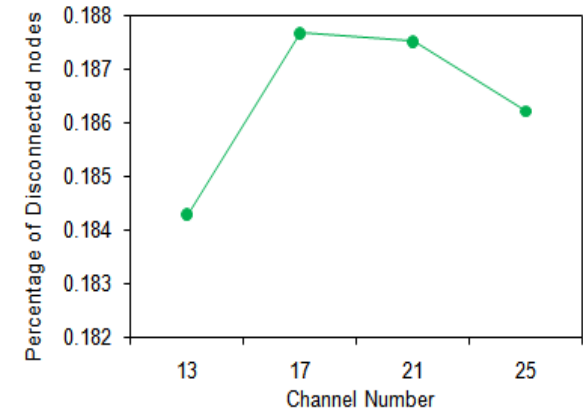
Latency with time vs. frequency



expected % disconnected nodes after 10 years



% of disconnected nodes vs. frequency



Net-Centric AADL Extensions Proposed

- In the queue of AADL v2.0 proposals
- Multiplicity specification
 - To specify properties of groups of nodes
- Location specification
 - Ability to specify node location in AADL model
 - Ability to construct arbitrary deployment topologies
 - Standard statistical distribution for deployment
- Lossy buses specification
 - Ability to specify analytic wireless link quality models
 - Models based on empirical measurements

Conclusion
